TOPSOE

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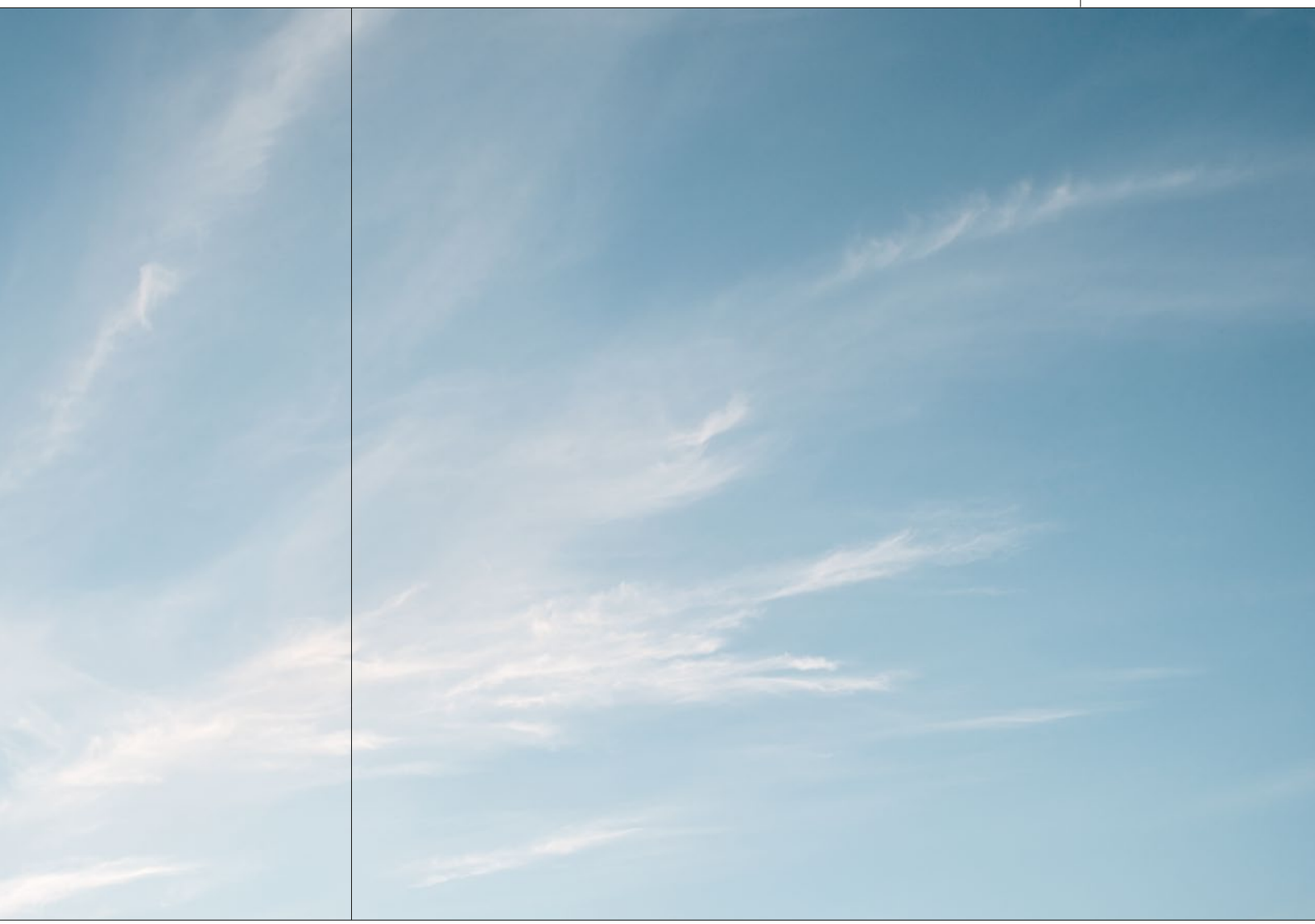
LET'S POWER UP REFORMING!

Electrify your low-carbon hydrogen production with **eREACT™** – the electrified counterpart to the world's most common hydrogen production method: steam methane reforming (SMR).



A RENEWABLE ROUTE FOR FOSSIL FEEDSTOCKS

The world is moving towards low-carbon production methods. Using renewable sources, such as wind and solar power, for electricity production will form the foundation for the energy systems of the future. This is great for our joint goal of reducing greenhouse gas emissions globally.

However, industries like shipping and aviation will likely rely on renewable and low-carbon fuels like hydrogen and its derivatives to achieve our net zero ambitions. More than 99% of global hydrogen production is still derived from fossil fuels, particularly natural gas which accounts for 76%[1].

To produce hydrogen from natural gas, steam methane reforming (SMR) is the most widespread route, and it is expected to remain the dominant process for hydrogen production because of its favorable economics and existing infrastructure.

In 2014, we at Topsoe embarked on a research project to leverage our existing knowledge and expertise in SMR technology to an emission-free alternative. Recognizing that the key to unlocking the potential for low-carbon hydrogen production lies in the electrified reforming technology used to produce the syngas, we devised a solution that seamlessly integrates electricity with reactor processes. This innovation enables a remarkable reduction of up to 99% in CO_2 emissions compared to conventional SMR.

We've named it **eREACT™**.

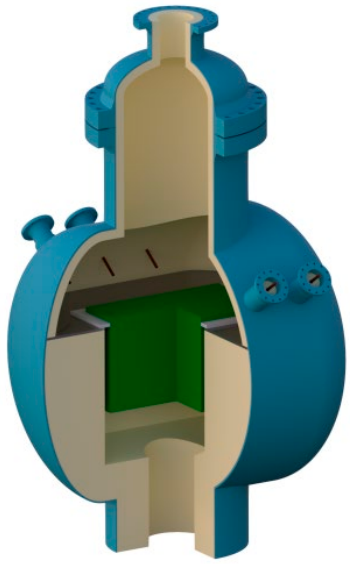
[1] IEA, Global Hydrogen Review 2025.

POWER UP LOW-CARBON HYDROGEN PRODUCTION

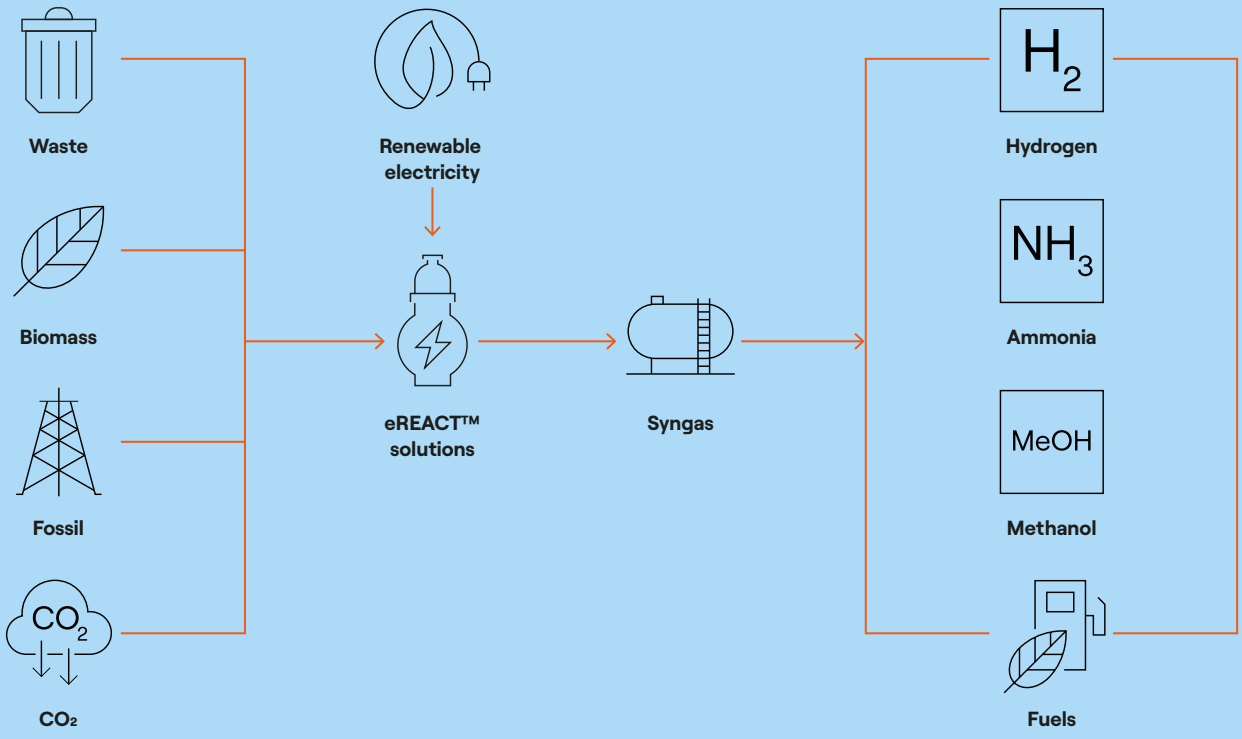
eREACT is an electrified alternative to steam methane reforming (SMR), the most widely used method for hydrogen production.

However, there is a significant difference between the two. While SMR relies on the combustion of natural gas for heat generation, resulting in CO_2 emissions, eREACT relies on (renewable) electricity to drive the same reaction, resulting in an ultra-low carbon intensity.

eREACT builds upon the thermodynamic principles utilized in fired steam methane reformers, but it offers an expanded operating window compared to conventional SMR technology. This allows for process intensification, including operation at higher temperatures and more precise and rapid control over the reactor. As a result, eREACT delivers exceptional carbon utilization, energy efficiency, and operational stability.



eREACT will empower you to future-proof your low-carbon hydrogen production with next-generation steam methane reforming.

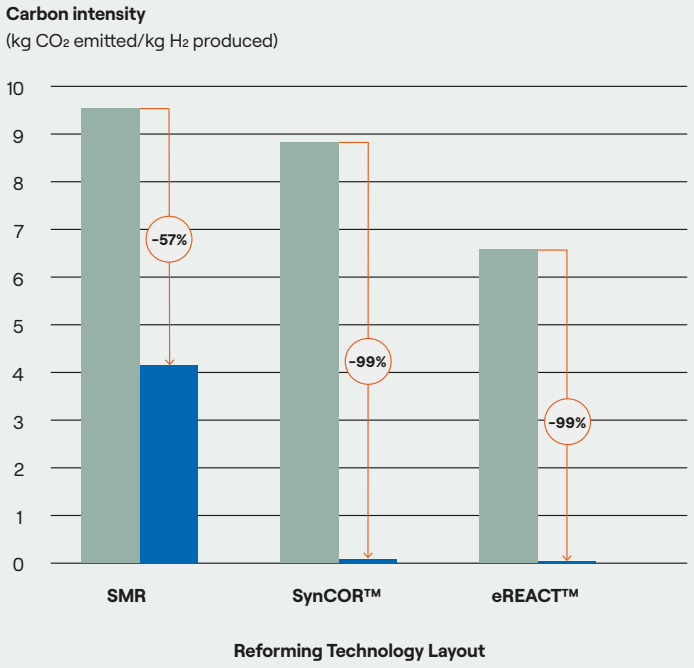


A CUTTING-EDGE CARBON CRUNCHER

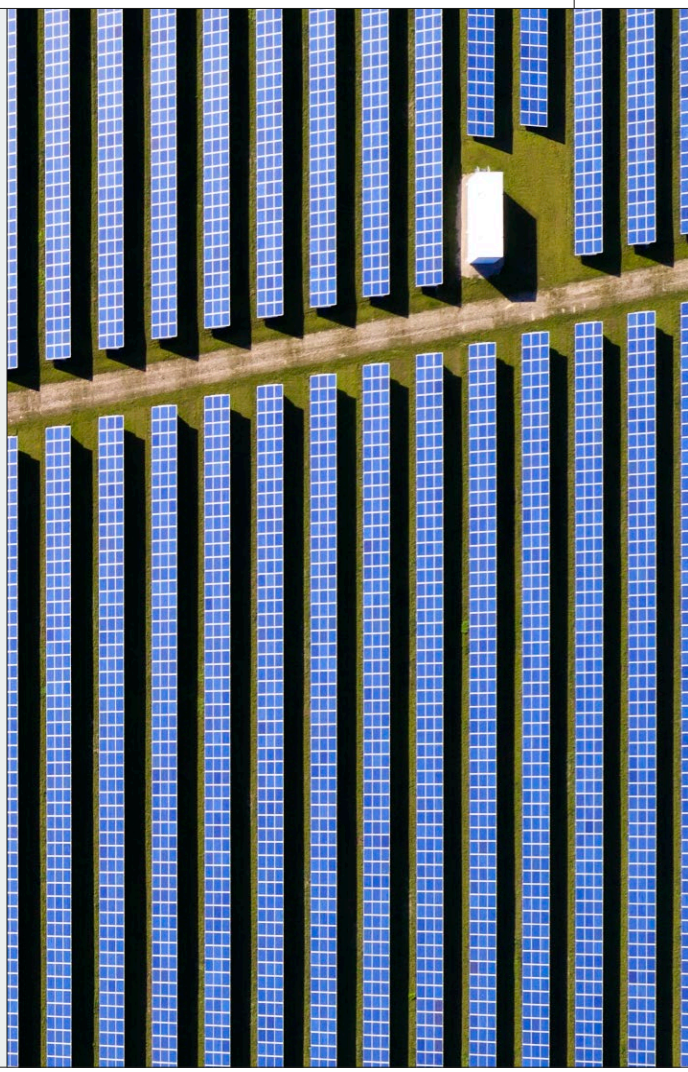
A key benefit of eREACT compared to conventional SMR is that it does not produce flue gas and the associated environmental consequences. This is achieved by generating reaction heat from renewable electricity directly in the reforming reactor.

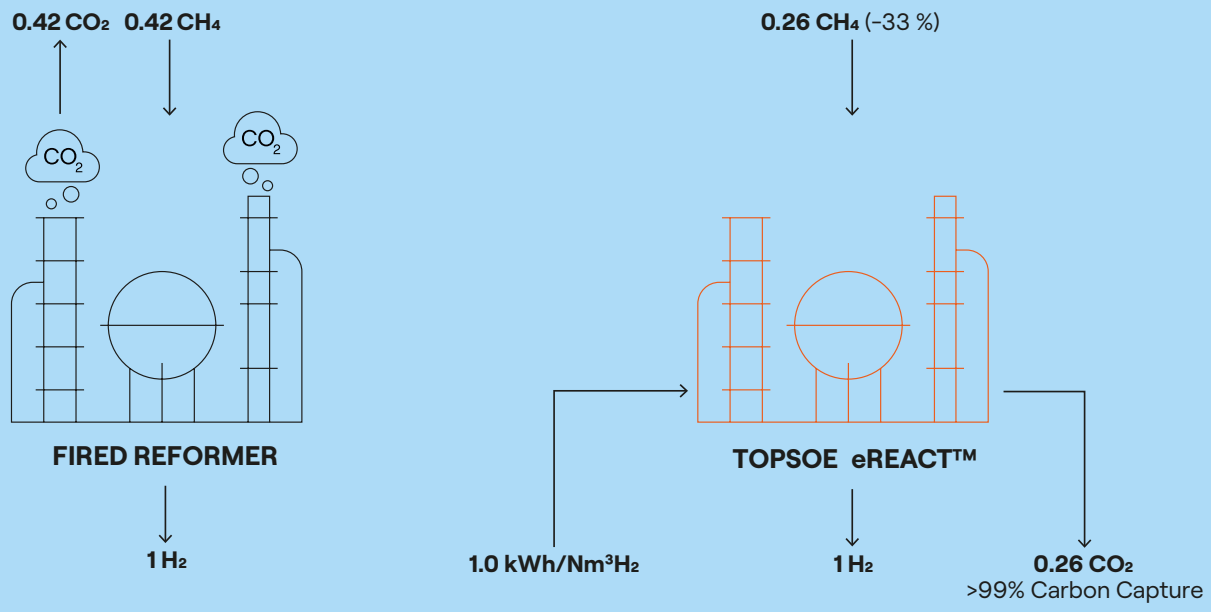
By incorporating renewable electricity into hydrogen production, you can use close to carbon neutral (more than 99%) hydrogen production from feedstock to syngas.

CARBON INTENSITY OF HYDROGEN PRODUCTION TECHNOLOGIES



- Carbon intensity (no CO_2 capture)
- Carbon intensity (with process CO_2 capture - without flue gas capture)





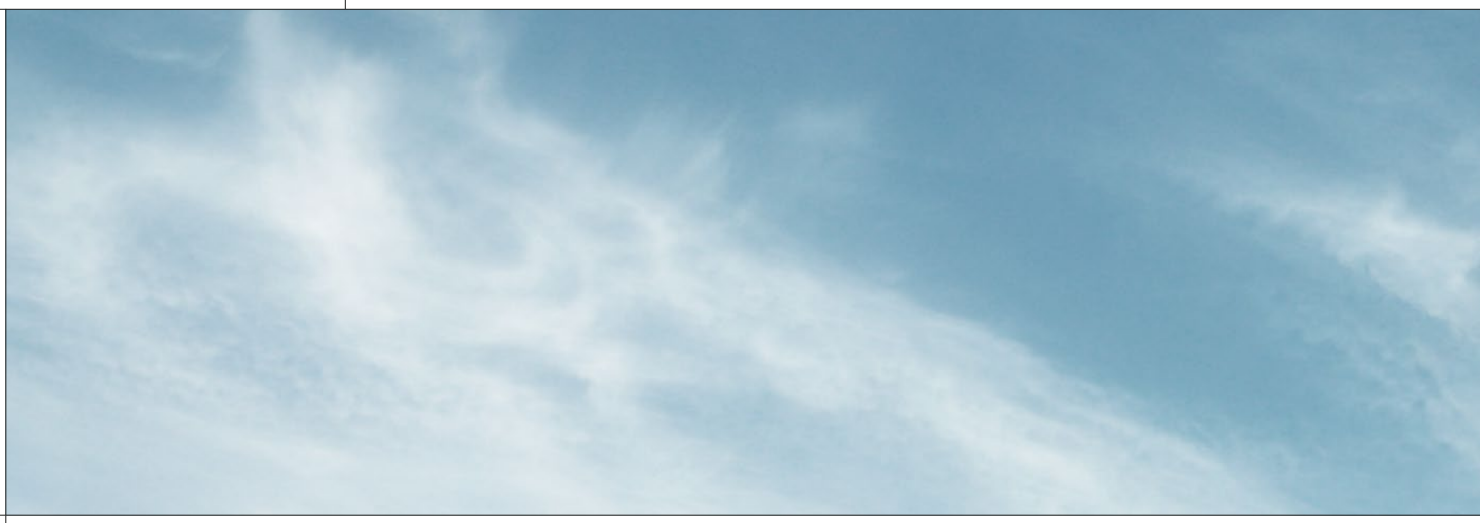
ELEVATING ENERGY EFFICIENCY
eREACT offers a significant advantage in energy efficiency, enabling a 30-40% reduction in natural gas consumption for hydrogen production compared to conventional SMR, which is achieved through the electrification of the heat generation. This makes eREACT a highly efficient electric reforming technology.

HARNESSING EXISTING PLANT INFRASTRUCTURE
eREACT allows existing industrial complexes to cost-effectively electrify their hydrogen production. The technology can be integrated into an existing production facility through a straightforward revamp of the hydrogen plant. Its modular design and streamlined equipment requirements make on-site construction simple and reduce general engineering costs. Additionally, eREACT enables decentralized production, further enhancing ease of implementation and operational efficiency.

The technology also benefits from safe and low-cost continuous operation and long-term ease of maintenance.

eREACT'S CUTTING-EDGE PERKS

- Reduces natural gas consumption with 30-40% compared to conventional SMR for hydrogen production
- CO_2 emissions reduction by up to >99% with eREACT compared to conventional SMR
- Ultra-low carbon intensity of 0-0.2 kg CO_2/kg H_2
- Eliminates all flue gas from the hydrogen-production process
- Lower engineering and operating cost through simpler plant design
- Safe and low-cost continuous operation
- Easy on-site construction and transportation
- Fits with decentralized chemicals production
- Faster startup time
- Long-term ease of maintenance

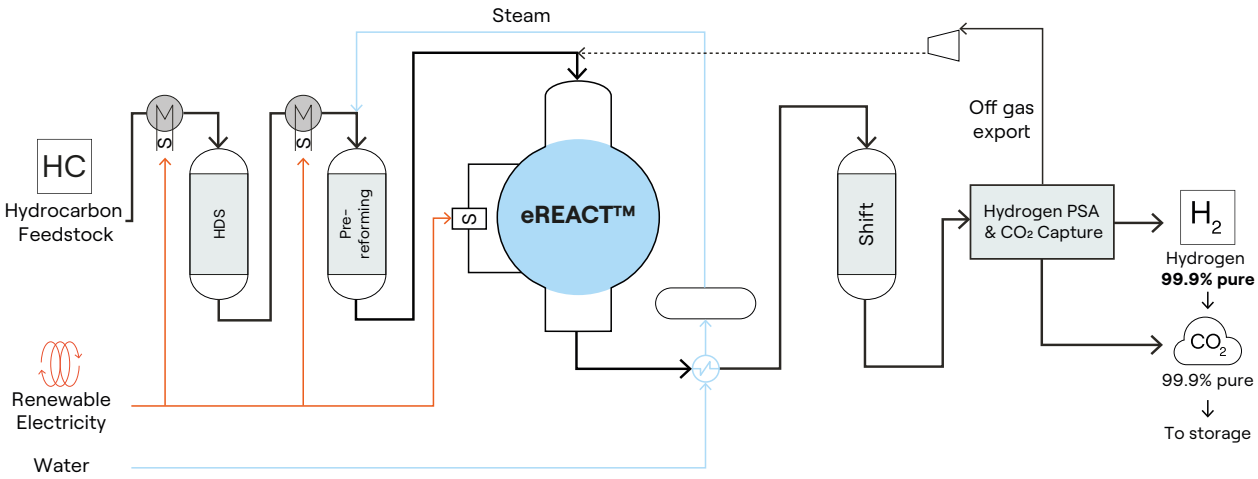


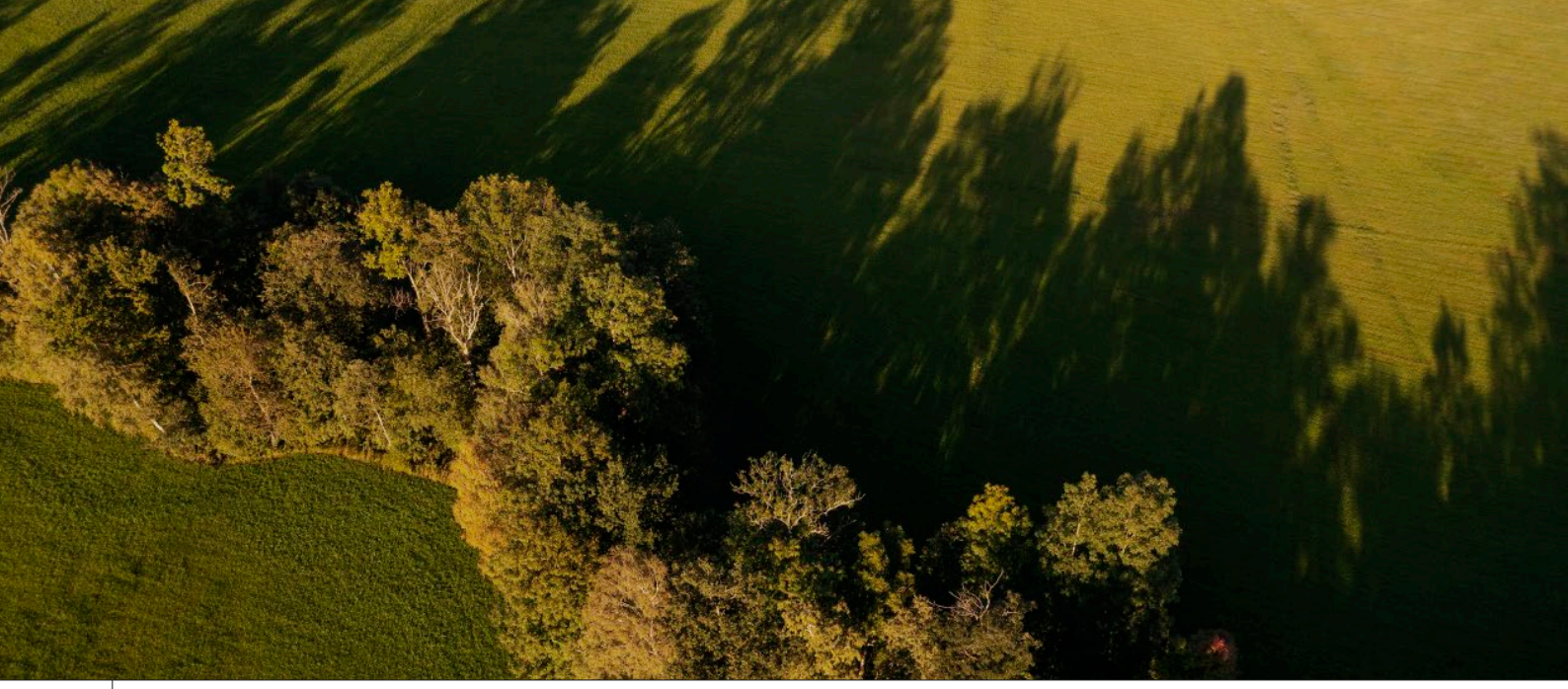
A SNEAK PEEK ON THE eREACT PROCESS

The hydrocarbon feedstock (such as natural gas, LPG, or naphtha) undergoes desulphurization before it's combined with steam. This mixed gas is efficiently preheated using electrical heating, and then enters the prereformer for a partial reforming of the gas. The prereformed gas is fed to the electrical reactor in which the methane is converted to synthesis gas by the steam reforming reaction. The compact reactor of eREACT features a catalyst bed designed and manufactured by Topsoe and requires an electricity input of approximately 1 kWh/Nm^3 of hydrogen. Finally, the gas is cooled down in a waste heat boiler.. This cooling process generates steam, which is used for the reforming process.

The final syngas is treated based on its intended application. For hydrogen production, the syngas is directed to a shift reactor, where it's purified through a hydrogen PSA (pressure swing adsorption) unit, and a CO_2 capture unit. This yields high-purity hydrogen.

FROM NATURAL GAS (OR OTHER HYDROCARBON FEEDSTOCKS) TO SYNGAS





4 KEY ELEMENTS OF THE eREACT DESIGN

1.

eREACT is an integrated system where heating is achieved through direct electrical resistance heating of the catalyst. This makes eREACT highly adaptable in operation, facilitating rapid start-up, capacity changes, and precise temperature control. Plant control is simplified to a direct feedback loop between syngas temperature and power supply unit (PSU). A PSU can adjust power levels within milliseconds.

2.

eREACT is constructed to have the hottest part centrally in the reactor. This enables a high-temperature operation at up to 1000°C. This allows for high methane conversion and potential operation at pressures exceeding 50 bar for innovative process integrations. This means improved feedstock utilization compared to conventional SMR-fired reformers, indirectly promoting more sustainable chemical production.

3.

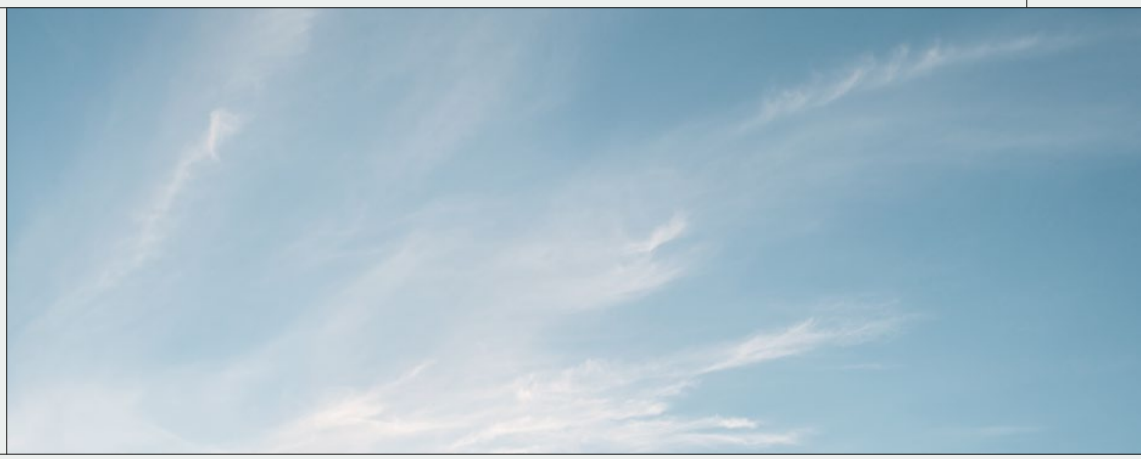
eREACT is less complex, resulting in a corresponding simplification in the layout of the chemical site. It is a more streamlined reactor compared to SMR, which significantly reduces the unit's volume. Additionally, the utility site experiences a reduction in complexity.

4.

eREACT offers the opportunity to retrofit existing syngas installations by integration with existing reforming units for increased syngas manufacturing, but where the added capacity does not have an associated CO_2 emission.

A PROVEN SOLUTION READY FOR DEPLOYMENT

For more than 10 years, Topsoe has researched and developed ways to transfer existing knowledge on SMR technology to an emission-free electrified counterpart. Having gone through in-house scale-up, eREACT is now a proven technology ready to be deployed, and its feasibility and scalability have been demonstrated through our pilot project.



The eREACT™ pilot plant unit in Denmark

eREACT PILOT PLANT IN DENMARK

In 2022, we installed a pilot plant at Aarhus University's research facility in Denmark (Foulum). Since its commissioning in October 2022, the plant has successfully demonstrated operational experience with the industrial-scale electrified steam methane reforming technology (eREACT). The plant has received funding from EUDP – The Danish Energy Technology Development and Demonstration Programme.

PARTNER WITH THE WORLD'S BIGGEST HYDROGEN LICENSOR

Topsoe is one of the world's biggest hydrogen licensors. This gives us a profound understanding of reforming technologies and integrations for low-carbon hydrogen production, and a unique understanding of the full value chain.

Our extensive experience in process design and catalysts is built on R&D and references on low-carbon hydrogen projects.

ABOUT TOPSOE

Topsoe is a leading global provider of technology and solutions for the energy transition. We combat climate change by helping our customers and partners achieve their decarbonization and emission reduction targets in a most optimal manner.

Based on decades of scientific research and innovation, we offer world leading solutions for transforming renewable resources into fuels and chemicals for a sustainable world, and for efficient and low-carbon fuel production and clean air.

We are founded in 1940 and headquartered in Denmark with over 2,800 employees serving customers all around the globe.

To learn more, visit **topsoe.com**

TOPSOE
Making Energy Transition

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